

Q3-A-S1

Question:

1. Given the following -

Processes	Base Register	Limit Register	Logical Address from the program	Corresponding physical address	Valid?
P1	16384	2048	30		
P2	24000	4096	3800		
P3	30000	1024	1200		

- a. For each process, calculate and fill up the column "Corresponding physical address".
- b. Fill-up "Valid?" column also by determine whether the memory access is valid or invalid for each process using the condition: Valid if Logical Address < Limit Register

Q3-B-S1:

Question:

Parameter	Value
Virtual address space	15 bits
Page size	512 bytes
Physical memory size	4 KB

- a. Calculate the number of page frames in physical memory.
- b. If a process generates a virtual address = 2000, find the corresponding physical address, assuming: Page 0 → Frame 3, Page 2 → Frame 0, Page 3 → Frame 5, Page 4 → Frame 2.
- c. What happens if the process tries to access an address that belongs to page 1?

Segmentation:

Given Segment Table:

Segment #	Base Address	Length (Limit)
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

Basic Translation

a) (0, 430)

Segmentation with paging:

Seg #	PT Base	Length (pages)
0	100	4
1	200	3
2	300	2

Page Tables:

- Seg 0: P0→F5, P1→F8, P2→F2, P3→F7
- Seg 1: P0→F10, P1→F3, P2→F9
- Seg 2: P0→F6, P1→F4

Assume page size = 512 bytes

a) (0, 2, 100)